

chapter 1

optical fiber communication system is the communication system based on fiber cable use for data transmission

General system:

It is often achieved by modulating a carrier, which, carries signal.

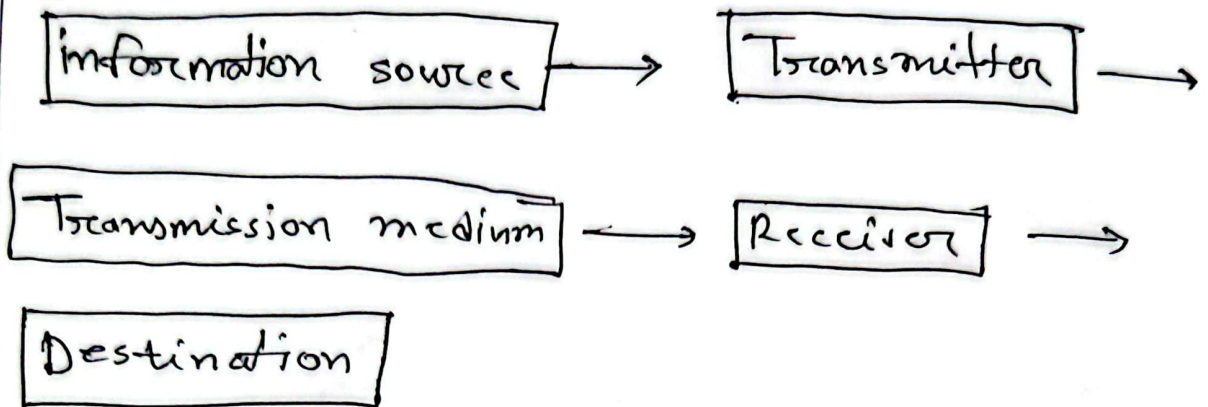
- A pair of wires
- A co-axial cable
- radio link

It is in any transmission medium signal is

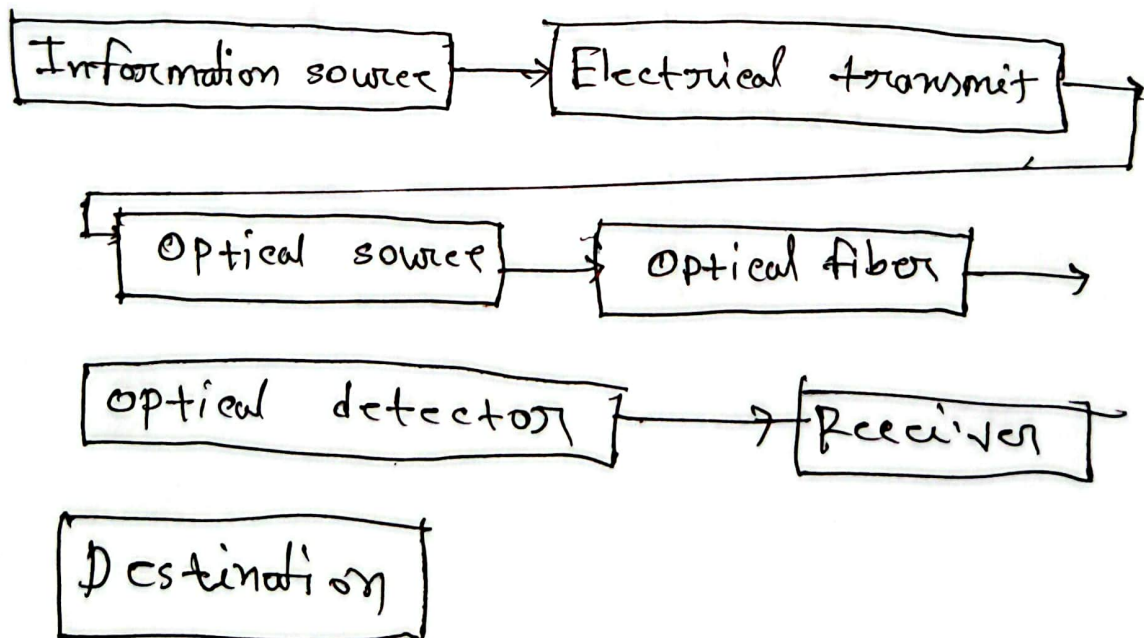
- attenuated
- suffers loss.

General communication system

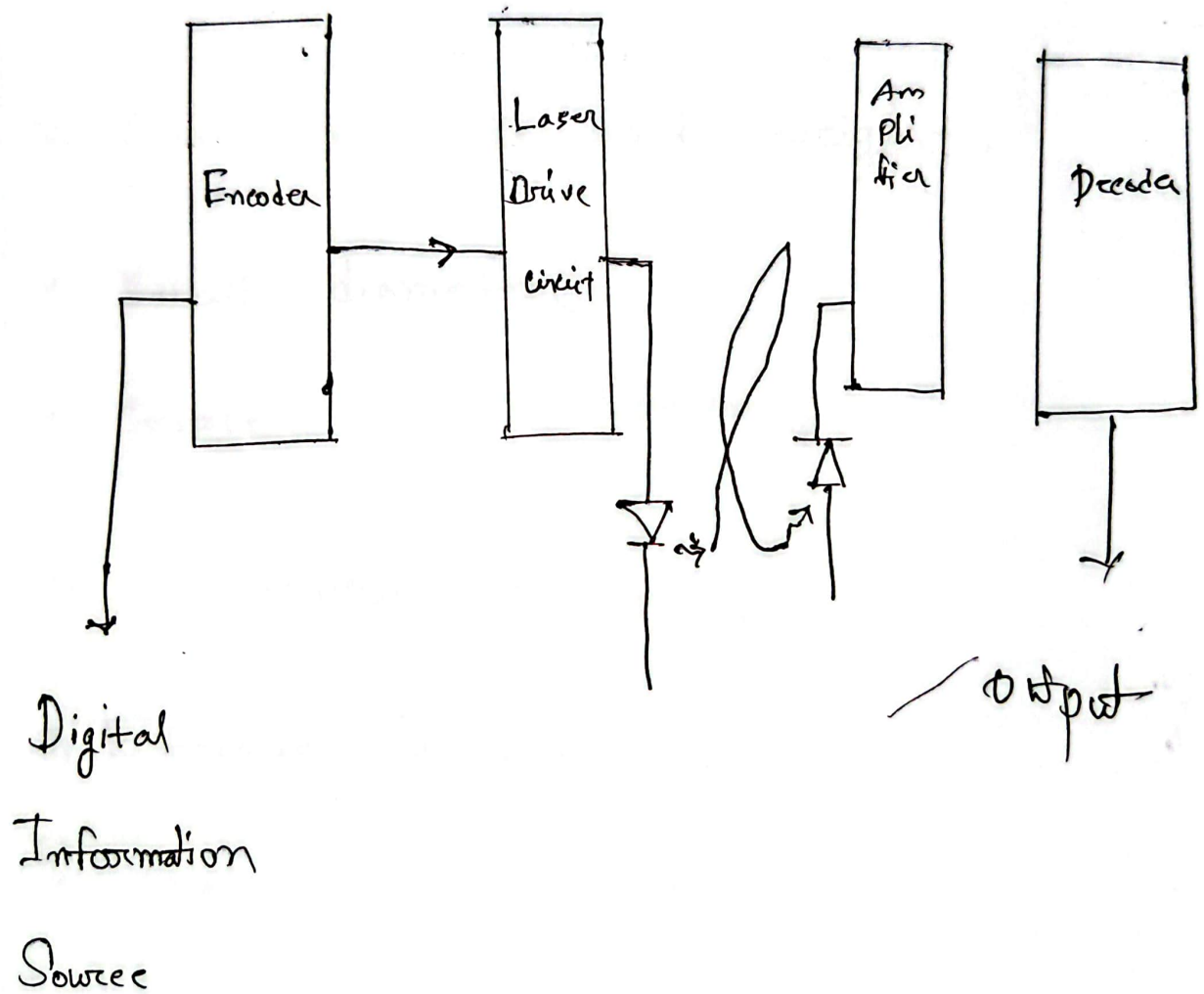
a)



Electrical communication



Optical fiber communication using semiconductor laser



Advantages of optical fiber

■ Enormous potential bandwidth

- Rang $10^{13} \rightarrow 10^{16}$ Hz
- Coaxial cable band 20 MHz

■ Small in size and weight.

- Small diameter
- Covered with coating
- Tremendous advantage

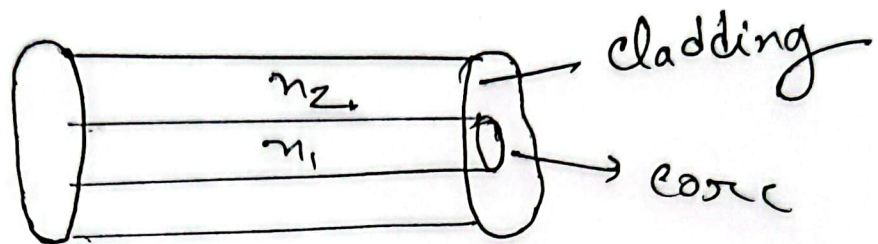
■ Electric Isolation

■ Immunity to interference

?

chapter - 2

Optical fiber waveguides

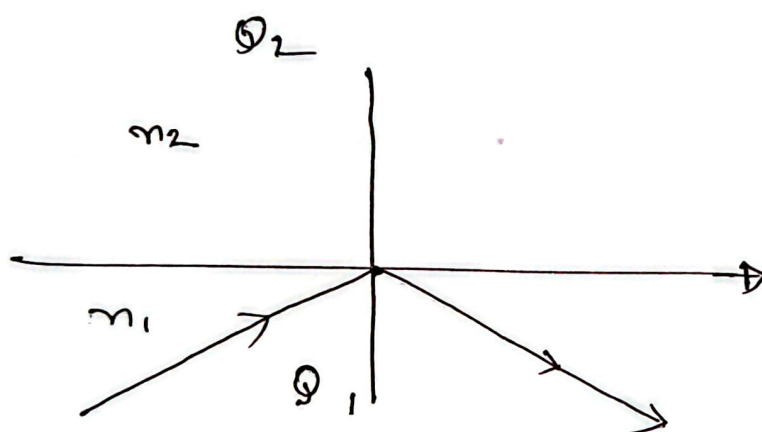
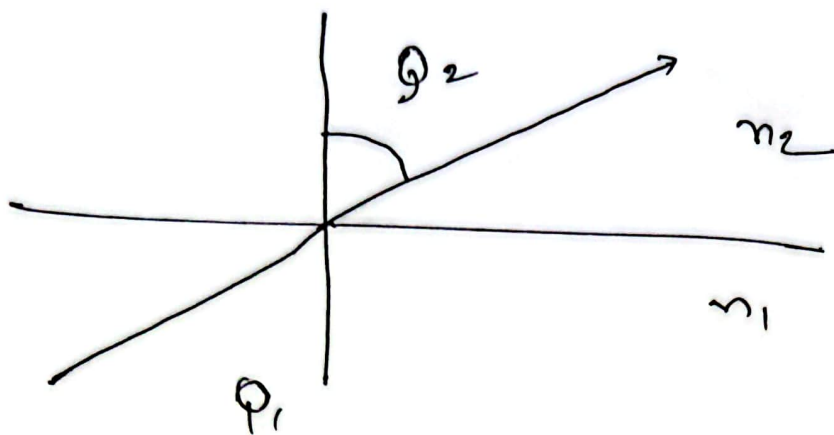


Total internal Reflection

Angle of incidence θ_1 , Refraction θ_2 .

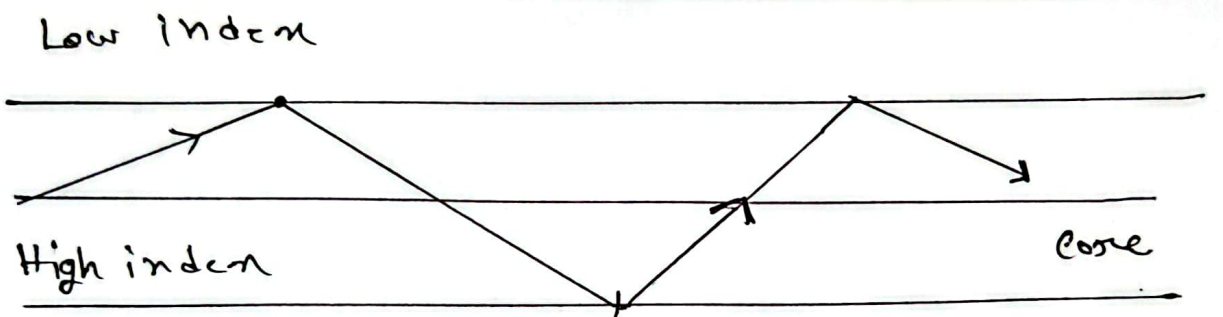
$$\therefore n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1}$$

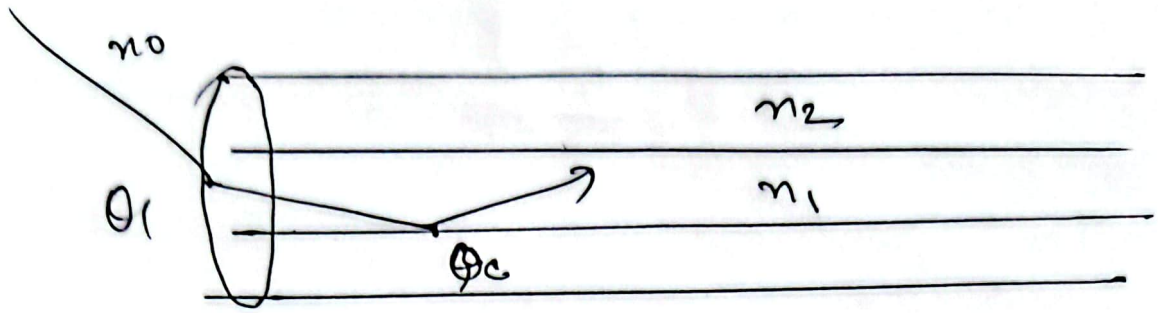


Total internal reflection

$$\sin \phi_c = \frac{n_2}{n_1}$$



The efficiency is around 99.9%.



$$n_0 \sin \theta_1 = n_1 \sin \theta_2$$

$$\phi = \frac{\pi}{2} - \theta_2$$

$$n_0 \sin \theta_1 = n_1 \sin \left(\frac{\pi}{2} - \theta \right)$$

$$= n_1 \cos \phi$$

$$= n_1 \sqrt{1 - \sin^2 \phi}$$

$$n_0 \sin \theta_1 = n_1 \sqrt{1 - \sin^2 \phi}$$

$$= n_1 \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

$$= \sqrt{n_1^2 - n_2^2}$$

$\therefore NA = \text{numerical aperture}$

$$= n_0 \sin \theta_a = \sqrt{n_1^2 - n_2^2}$$

$n_0 = 1$ for air

$$\sin \theta_a = \sqrt{n_1^2 - n_2^2}$$

$$\begin{aligned} \Delta &= \frac{n_1^2 - n_2^2}{2n_1^2} \\ &= \frac{(n_1 n_2)(n_1 + n_2)}{2n_1^2} \\ &= n_2 \end{aligned}$$

$$2\Delta n_1^{\vee} = n_1^{\vee} - n_2^{\vee}$$

$$\Rightarrow \sqrt{n_1^{\vee} - n_2^{\vee}} = \sqrt{2\Delta n_1^{\vee}}$$

$$NA = n_1 \sqrt{2\Delta}$$

$$\therefore n_0 \sin \theta_{\text{as}} \cos \gamma = \sqrt{n_1^{\vee} - n_2^{\vee}} = NA$$

$$\sin \theta_{\text{as}} = \frac{NA}{\cos \gamma}$$

$$\therefore \theta_{\text{as}} = \sin^{-1} \left(\frac{NA}{\cos \gamma} \right)$$

chapter 3

- Attenuation : Attenuation is the transmission loss.

It is measured by

$$(dB) = 10 \log \frac{P_i}{P_o}$$

$$\text{Per length attenuation } \alpha L = 10 \log \frac{P_i}{P_o}$$

Attenuation factors :

- material absorption
- material scattering
- Microbending
- mode coupling
- Leaky modes,

$$\bullet A = \frac{n_1^{\sqrt{}} - n_2^{\sqrt{}}}{2n_1^{\sqrt{}}}$$

$$\bullet R_e = \frac{3n_1^{\sqrt{}} d}{4\pi (n_1^{\sqrt{}} - n_2^{\sqrt{}})^{3/2}}$$

$$\bullet \lambda_c = \frac{217 \times \alpha \times n_1 (24)^{1/2}}{2.405}$$

• photocurrent I_p : actual electrical signal that will be amplified and processed by the rest of the computer system to recover the transmitted data

$$p = \frac{nq}{h\nu} = \frac{nqd}{hc}$$

Signal dispersion

- Material dispersion : Because of Refractive index fluctuation
- pulse broadening : For different speed and time of light.
- Intersymbol interference : Overlap of pulse ,

pulse broadening $\Delta T = D \times L \times \Delta \lambda$

$$\text{Bit rate} = \frac{1}{2\Delta T}$$

Passive components

1. Mux and demux : splitting and integrating signal.
2. Optical switches : input output power controlling
3. wavelength converter : Inter changing wavelength.
4. coupler : split or combine signal
5. circulator and isolator : one direction allowance.
6. Attenuator : Reduction process of power
7. Optical filter : filters the signals for specific target signal.